Orders, Symmetries, Products, and Categorical Cancellation

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An arrow in a category need not be a function. In a partially ordered set it records a comparison; in a group it records a reversible operation. The real test is whether concepts such as product, injection, and surjection can be recovered entirely from arrows. The examples below follow Tom Leinster's *Basic Category Theory*, Chapters 1–2.[1]

§1.1 From reflexivity and transitivity to composition

Let P be a preorder: its relation $\preceq$ is reflexive and transitive. Define

$$\mathrm{Hom}(p,q) = \begin{cases} \{*\}, & p \preceq q, \\ \varnothing, & p \not\preceq q. \end{cases}$$

Reflexivity supplies the identity $p \to p$; transitivity supplies composition. Associativity follows automatically because there is at most one arrow with each source and target.

For example, take the positive divisors of 12, ordered by divisibility. The arrows $2 \to 6$ and $6 \to 12$ compose to $2 \to 12$. A Hasse diagram shows only the generating cover relations; the corresponding category contains all composites as well.

An isomorphism $p \cong q$ requires both $p \preceq q$ and $q \preceq p$. If the preorder is a poset, antisymmetry forces $p = q$. In a general preorder distinct isomorphic objects are possible. Thus a poset category is a special, skeletal form of a preorder category.

§1.2 A group as a one-object category

Given a monoid M, construct $\mathcal{B}M$ with just one object $*$ and

$$\mathrm{Hom}_{\mathcal{B}M}(*,*) = M.$$

Multiplication is composition, and the unit is the identity arrow. These are exactly the associative and unit axioms of a category. If M is a group, each arrow is invertible, making $\mathcal{B}M$ a one-object groupoid.

This also translates group actions into the language of functors. A functor

$$F : \mathcal{B}G \longrightarrow \mathbf{Set}$$

chooses a set $X = F(*)$ and a map $F(g) : X \to X$ for each $g \in G$, satisfying

$$F(gh) = F(g) \circ F(h), \qquad F(e) = \mathrm{id}_X\,.$$

Since g has an inverse, $F(g)$ is a bijection. These conditions are precisely a group action. The functor axioms are doing the mathematical work.

[1] https://arxiv.org/abs/1612.09375

§1.3 Products of categories versus products of objects

In $\mathcal{C} \times \mathcal{D}$, objects are pairs (C, D), arrows are pairs (f, g), and composition is componentwise. There are projection functors

$$\pi_{\mathcal{C}} : \mathcal{C} \times \mathcal{D} \to \mathcal{C}, \qquad \pi_{\mathcal{D}} : \mathcal{C} \times \mathcal{D} \to \mathcal{D}.$$

For any $\mathcal{E}$, a functor $H : \mathcal{E} \to \mathcal{C} \times \mathcal{D}$ is uniquely determined by its projections. Conversely two functors $F : \mathcal{E} \to \mathcal{C}$ and $G : \mathcal{E} \to \mathcal{D}$ define $H(e) = (F(e), G(e))$, with the analogous rule on arrows. This is the product's universal property in the category of small categories.

One must distinguish this from a product *inside* a category. In **Set**, $A \times B$ is a single set with two projection functions, and every pair $X \to A$, $X \to B$ factors through it uniquely. Both constructions satisfy similar diagrams, but their objects live at different levels.

§1.4 Opposite categories and the direction of a test

The opposite category $\mathcal{C}^{\mathrm{op}}$ reverses all arrows, with composition reversed accordingly. The divisor-poset category becomes reverse divisibility; a terminal object becomes initial. Statements phrased entirely by morphisms admit dual statements upon reversing their arrows.

This motivates a definition without reference to individual elements.

Definition 1.4.1. A morphism $f : A \to B$ is *monic* if

$$fg = fh \Longrightarrow g = h \qquad (g, h : X \to A),$$

and *epic* if

$$gf = hf \Longrightarrow g = h \qquad (g, h : B \to X).$$

§1.5 Recovering injection and surjection in sets

In **Set**, monic is equivalent to injective. An injective map cancels pointwise. Conversely, if $f(a) = f(b)$, consider the singleton $\{*\}$ with maps $g(*) = a$, $h(*) = b$. Monicity of f implies $g = h$, hence $a = b$.

Similarly, surjections are epic: agreement of two maps after a surjection forces them to agree everywhere. If $f : A \to B$ omits $b \in B$, define two functions $g, h : B \to \{0, 1\}$ that agree on $f(A)$ and disagree only at b. They satisfy $gf = hf$ but $g \neq h$, so f cannot be epic.

The singleton and the two-point set act as *test objects*. This calculation proves the agreement with familiar set theory, rather than assuming that a category can discuss elements.

§1.6 Two ways the analogy can fail

In a poset category all arrows are simultaneously monic and epic: any two parallel arrows are equal automatically, so both cancellation implications hold. Yet $2 \to 6$ in the divisor-poset of 12 is not invertible, because $6 \nmid 2$.

A less trivial example occurs among commutative unital rings: the inclusion

$$\iota : \mathbb{Z} \longrightarrow \mathbb{Q}$$

is epic without being surjective as a function. If ring homomorphisms $g, h : \mathbb{Q} \to R$ agree on $\mathbb{Z}$, they agree on every $a/b \in \mathbb{Q}$. Indeed $g(b) = h(b)$ is a unit, whose inverse is unique, so

$$g(a/b) = g(a)g(b)^{-1} = h(a)h(b)^{-1} = h(a/b).$$

Thus $g = h$. If no such homomorphisms exist, the cancellation condition is vacuous.

The lesson is a mathematical distinction: categorical cancellation is intrinsic to composition; injectivity and surjectivity refer to an underlying set and can change meaning across categories. The reason to study monomorphisms and epimorphisms is precisely that the arrow test still makes sense when the elements do not.